CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

M.Tech. (CAD/CAM) Sem - I University Examination, 2010-11

CE – 725 Artificial Intelligence

Date: 12.05.11, Thursday Time: 10:00 a.m To 01:00 p.m Maximum Marks: 70

Instructions: (1) All questions are compulsory.
(2) Indicate clearly, the option you attempt along with its respective question number.
(3) Figures to right indicate marks
(4) Rough work is done in the last page of main answer book, don't write anything on the question paper.

SECTION-I

Q-1 **Answer the following questions.**

1. What are the components of Artificial Intelligence? Explain production system in brief. **3**
2. Discuss the important characteristics of production system and control strategy. **3**
3. Justify with reason: Backtracking is not a solution to local minima problem. **2**
4. What are prons and cons of BFS and DFS? **3**

Q-2

[A] Devise good heuristic function for 8- puzzle problem and show that how it can solve any 8-puzzle problem. **4**

[B] Write a program to add the element at the head and tail. **4**
e.g. ([a, b, c] [d, a, b, c, d]).

[C] Draw the semantic net for following sentence: **4**
"Every batter hits a ball."

OR

[A] Explain: Multilayer Feed forward Networks. **4**

[B] What are the attractive features of an Expert System? Compare Expert System with traditional software. **4**

[C] Explain procedural knowledge and Declarative knowledge with example. **4**

Q-3

[A] Convert the following sentences into Predicate Logic and convert it into Clausal Form. **4**

1. Some airplanes are faster than every airplane.
2. If an integer is not even, then it is odd.
3. Every apple is either green or yellow.
4. There is some table that doesn't have 4 legs.

[B] Do the following sentences unify? If yes, give unifier. **4**
Knows (john,x)
Knows (y, mother(y))
Knows(x), knows (g(y)).

[C] Discuss how First Order Predicate Logic (FOPL) is powerful than proposition logic? What are the limitations of proposition logic? **4**

OR

[A] Write a prolog program to find out maximum and minimum of two numbers, three numbers. **4**

[B] Give good heuristic for Blocks World Problem and show that how it can solve any sample block world problem. **4**

[C] Write a PROLOG program to generate family tree. **4**

SECTION-II

Q-4

1. What are the problems with processing of English language? **3**
2. Compare Nonhierarchical Planning with Hierarchical Planning. **4**
3. Give Prons and Cons of Resolution Proof Method. **4**

Q-5

[A] Define following with respect to algorithm performance. **4**

1. Admissibility
2. Branching Factor

[B] What are the applications of neural networks? What do you mean by Hopfield Network? What are the major limitations of it? **4**

[C] Compare Fuzzy logic with Binary logic. **4**

OR

[A] What are the applications of an Expert System? Who are the users of an Expert System? **4**

[B] Which are the factors influencing Backpropagation Neural Network Training? **4**

[C] Why NLP is required? Explain the phases of NLP. **4**

Q-6

[A] Derive the parse tree for the sentence using the appropriate rules: "John has deleted .init file". **4**

[B] How Bayesian Belief Nets are used to classify items for a given problem? Write a short note on Bayesian Belief Networks. **4**

OR

[B] Solve the following crypt arithmetic problem. **4**

LOGIC
+LOGIC
PROLOG

[C] What are seven problem characteristics? **4**

OR

[C] Differentiate between Cut and Fail predicates with example. **4**